



passwd

DEEP DIVE

Prepared By: Hosni Zaaraoui

What is passwd?

The `passwd` command is used to set or change user passwords in Linux. It also allows administrators to control account aging policies, lock/unlock accounts, and enforce password policies.

Basic Syntax

`passwd [OPTIONS] [USERNAME]`

If no `USERNAME` is provided, it modifies the password for the current user.

Common Use-Cases and Options

Change Your Own Password

passwd

- Prompts the current user to change their password.
- Requires the old password before setting the new one.
- Enforces system password policies (length, complexity, history, etc.)

```
hosni ~ 09:35$ passwd  
Changing password for hosni.  
Current password:
```

Common Use-Cases and Options

Root Changing Another User's Password

sudo passwd username

- Used by root to reset another user's password.
- Does not require the user's current password.

```
hosni ~ 09:59 $ sudo passwd ahmed
New password:
Retype new password:
passwd: password updated successfully
```

Common Use-Cases and Options

Lock a User Account

sudo passwd -l username

- Adds a ! in /etc/shadow to disable login with a password.
- Account can still be accessed via public key (if configured).

```
hosni ~ 10:01 su ahmed
Password:
$ exit

hosni ~ 10:02 sudo passwd -l ahmed
passwd: password expiry information changed.

hosni ~ 10:02 su ahmed
Password:
su: Authentication failure
```

Common Use-Cases and Options

Unlock a Locked Account

`sudo passwd -u username`

- Removes the ! from the password field in `/etc/shadow`.

```
hosni ~ 10:02 sudo passwd -u ahmed  
passwd: password expiry information changed.
```

```
hosni ~ 10:03 su ahmed  
Password:  
$ whoami  
ahmed  
$ _
```

Common Use-Cases and Options

Set Password Expiry Info

sudo passwd -x 30 username

- Sets password to expire after 30 days.

Other related options:

- **-x DAYS:** Maximum number of days before a password change is required.
- **-n DAYS:** Minimum number of days between password changes.
- **-w DAYS:** Number of days before expiration to warn the user.
- **-i DAYS:** Number of inactive days after password expires before account is locked.

Common Use-Cases and Options

Expire a Password Immediately

sudo passwd -e username

- Forces user to change their password at the next login.

```
hosni ~ 10:03 sudo passwd -e ahmed  
passwd: password expiry information changed.
```

```
hosni ~ 10:04 su ahmed
```

```
Password:
```

```
You are required to change your password immediately (administrator enforced).
```

```
Changing password for ahmed.
```

```
Current password: _
```

Common Use-Cases and Options

View Password Status

sudo passwd -S username

Shows password status:

- P = Password set
- L = Account locked
- NP = No password

```
hosni ~ 10:04 sudo passwd -S ahmed  
ahmed P 01/01/1970 0 99999 7 -1
```

```
hosni ~ 10:04 sudo passwd -l ahmed  
passwd: password expiry information changed.
```

```
hosni ~ 10:05 sudo passwd -S ahmed  
ahmed L 01/01/1970 0 99999 7 -1
```

Common Use-Cases and Options

Remove User Password

sudo passwd -d username

- Deletes the user's password.
- User can login without a password (if allowed by PAM policy).

```
hosni ~ 10:07 $ sudo passwd -d ahmed  
passwd: password expiry information changed.
```

```
hosni ~ 10:07 $ su ahmed  
$ _
```

Real-World Scenarios

Scenario 1: Force Secure Password Change Policy

`sudo passwd -x 60 -n 3 -w 10 devops`

- Deletes the user's password.
- Password valid for 60 days
- Must wait 3 days before changing again
- Warn user 10 days before expiry

Real-World Scenarios

Scenario 2: Disable Temporary Developer Account

sudo passwd -l tempdev

- Prevents login without deleting the user.

Scenario 3: Reset Password After Suspicious Activity

sudo passwd suspicioususer

sudo passwd -e suspicioususer

- Admin resets password and forces change at next login.

Real-World Scenarios

Scenario 4: Audit Account Status Quickly

sudo passwd -Sa

- Check password status for all users.

```
pulse L 09/26/2023 0 99999 7 -1
hplip L 09/26/2023 0 99999 7 -1
hosni P 08/08/2024 0 99999 7 -1
swtpm L 08/09/2024 0 99999 7 -1
libvirt-qemu L 08/09/2024 0 99999 7 -1
libvirt-dnsmasq L 08/09/2024 0 99999 7 -1
ftp L 08/12/2024 0 99999 7 -1
whoopsie L 08/16/2024 0 99999 7 -1
gdm L 08/16/2024 0 99999 7 -1
sshd L 10/11/2024 0 99999 7 -1
jenkins L 10/11/2024 0 99999 7 -1
glances L 11/27/2024 0 99999 7 -1
vnstat L 11/30/2024 0 99999 7 -1
ahmed NP 08/04/2025 0 99999 7 -1
```

Option	Description
passwd	Change current user's password
passwd user	Change another user's password (as root)
-l	Lock account
-u	Unlock account
-e	Expire password
-d	Delete password
-S	Show account status
-n DAYS	Set min days before change
-x DAYS	Set max days password is valid
-w DAYS	Set warning days before expiry
-i DAYS	Set inactive days after expiry